



Towards a Meta-Modeling Approach for Social Business Process Requirements Capture

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ABSTRACT

A Social Business Process (SBP) is the result of blending social computing (*a.k.a.* Web 2.0) with business process (BP). Despite the benefits of SBP to enterprises, several limitations continue to undermine them. In this paper, we address two specific limitations, namely the difficulty of capturing SBP's requirements and the lack of a definition for SBP. Thus, meta-modeling is used to capture requirements from organizational, technological, and management perspectives. In addition, we introduce a definition for SBP by enriching an existing BP meta-model with social concepts. To annotate the SBP model with its requirements, a BPMN extension is proposed. The proposed meta-models are evaluated in terms of completeness and clarity using the Bunge-Wand-Weber ontology.

Keywords

Requirements; Social Business Process; Meta-Modeling; Bunge-Wand-Weber ontology

1. INTRODUCTION

With the recent boom in Web 2.0 technologies (e.g., JSON and AJAX) and applications (e.g., Facebook and Twitter), there is a major trend in blending social computing with other forms of computing like service computing and mobile computing [1, 30]. In this paper, we focus on enterprise computing where this blend results into a social business process (SBP) [12] and [36]. An SBP capitalizes on Web 2.0 technologies and applications to ensure that the social dimension of the enterprise (*a.k.a.* *Enterprise 2.0* [23]) is not overlooked neither at design nor run-time. Different studies have demonstrated the value-added of social computing to enterprises in terms of reaching out to more customers, collecting customers' online feedback, and profiling customers [9]. In addition, many enterprises recognize the need of rethinking

their strategies and reevaluating their operation models as the world is getting more "social" [7].

Despite the growing interest in SBP, several limitations continue to hinder this growth with emphasis, in this paper, on how to equip BP engineers with the necessary methods and techniques for capturing the SBP requirements. These requirements describe the necessary "*ingredients*" that would ensure an efficient adoption of SBPs in the enterprise. We also concentrate on the lack of a definition of an SBP.

In terms of solutions to these limitations, we first develop an approach for capturing socialization requirements from three perspectives: organizational, technological, and management. The organizational perspective deals with the necessary procedures that need to be put in place in order to ensure an efficient use of Web 2.0 technologies and applications. The technological perspective tackles the selection of the appropriate Web 2.0 technologies and applications that will help the enterprise sustain its growth and fulfill its mission. Finally, the management perspective is about the relevant metrics (or key performance indicators) used to evaluate the efficient use of Web 2.0 applications. Then, based on the identified requirements, we enrich the BP meta-model proposed in [10] with specific social details. We have chosen this meta-model due to its wide use in the research community for different purposes such as evaluation of BP modeling languages [22] and modeling flexible [6] and inter-organizational BPs [11]. The proposed SBP meta-model (SBP2M) serves to define what an SBP is? However, the lack of a definition of an SBP is an important issue that we address in this paper.

In the literature, there exist several approaches to capture requirements: goal-oriented [34], aspect-driven [27], use case [8], and meta-modeling [15]. The selection of an adequate approach depends on the goals of requirements specifying. In this paper, we aim to provide a means for capturing and reusing socialization requirements thanks to a Business Process Modeling Notation (BPMN) extension [24]. Thus, meta-modeling seems an adequate approach as BPMN extensibility mechanism is based on meta-models. In addition, meta-modeling promotes model reuse thanks to existing methods for software development like model-driven architecture [33]. Our approach sets the ground for developing different meta-models that will capture socialization requirements and allows the modeling of these requirements within a SBP model. The proposed meta-models are evaluated in terms of completeness and clarity using the Bunge-Wand-

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Weber ontology [35].

The remainder of this paper is structured as follows: Section 2 discusses related work on SBP development. Section 3 discusses the proposed meta-modeling approach. Section 4 defines SBP. Section 5 extends BPMN for modeling socialization requirements. Section 6 validates the proposed approach. Finally, Section 7 summarizes the contributions and highlights their extensions.

2. RELATED WORK

Brambilla et al. [3] propose a social extension of BPMN to model SBP. This extension defines new annotations that represent specific social network functionalities like posting, commenting, and invitation. In addition, Brambilla et al. present a technical framework based on MDA to transform an SBP model into a Web application integrated with public or private Web social networks. Despite, the importance of the developed framework for the adoption of SBP, it does not consider the SBP requirements during the different stages of SBP lifecycle.

In 2014, IBM proposed seven repeatable patterns that may be used to improve BPs by adding social capabilities [19]. Each social pattern is described in terms of: the concerned BPs, relevant industries, business challenges, key stakeholders, recommended actions, potential benefits as well as the IBM experience. The proposed patterns are applicable by organizations in various industrial sectors, independently of the enterprise particular context, e.g. its employees' readiness, organizational structure, policies, etc. In addition, the proposed patterns do not offer any guidelines for the capture of SBP requirements.

Rangiha et al. [17] propose a goal-oriented approach to define a Social Business Process Management (SBPM) system that uses social software to engage the final users in the process from modeling to enactment. This work addresses the lack of an effective method for applying social elements to BPM especially during the enactment stage. As such, it proposes to introduce goals starting at the modeling stage to act as controls. Thanks to the goals, the users can specify the sequence of tasks to be performed when executing the processes.

Cerenkovs et al. [5] propose an iterative, incremental development process, which aims to incorporate social interactions into BPs. This process consists of four phases: inception, elaboration, construction, and transition. The inception phase provides some recommendations that may help the enterprise assess BP socialization feasibility. However, authors did not propose any technical solution to support this identification phase. The second phase designs the SBP model using the BPMN standard. The social aspect is represented using Brambilla' extension for SBP modeling [3]. The third phase implements the SBP. Finally, the fourth phase allows the enactment of the new SBP. While the proposition of Cerenkovs et al. covers the main phases of the BP lifecycle, it informally specifies the four phases and it does not provide any information about the used techniques, standards, nor implementation of the process.

BP socialization research is at an early stage and there is still lack of an effective method for the adoption of SBP within organizations. Besides, we note the absence of an effective method for capturing SBP requirements. Such a method should help enterprises put in place the adequate procedures for an efficient use of Web 2.0, identify the most

adequate Web 2.0 application, and define the necessary metrics for SBP evaluation. We also note the lack of a definition for SBP.

3. META-MODELING FOR CAPTURING SOCIAL BUSINESS PROCESS REQUIREMENTS

In this section, we discuss the importance of dealing with SBP requirements from the organizational, technological, and management perspectives. Afterward, we present the meta-models we develop for capturing these requirements.

3.1 Requirement types

Socialization may lead into changes in entities like organizational unit, infrastructure, and management procedures [4, 26]. First of all, before the adoption of an SBP it is deemed necessary to guarantee a proper organizational structure in which an SBP along with its social tasks will be executed. To this end, we identify how to adopt the current organizational structure to SBP requirements. Furthermore, the execution of an SBP may require defining new business roles, duties, security policies, etc. To support these changes, we deal with SBP requirements from an organizational perspective. This allows to set up the necessary guidelines for an efficient use of the Web 2.0 technologies and applications.

In addition, the adoption of an SBP may require checking the infrastructure readiness. To assess this readiness, the relevant tools, services, and hardware resources should be identified. Therefore, we deal with requirements from a technological perspective, which provides a whole view of the IT components required for the execution of an SBP. These requirements help BP engineers find the most suitable Web 2.0 technologies and applications and predict socialization costs [32].

Finally, once adopted, an SBP must be evaluated to determine the value-added of socialization to the enterprise [32]. To this end, we define how to assess this value-added and measure the degree of achievement of the socialization goals. These goals indicate why the enterprise needs to use Web 2.0; for example to reduce communication or advertisement costs. The degree of achievement of socialization goals can be measured with key performance indicators [14]. The SBP requirements from a management perspective serves to provide for an adequate evaluation means for SBP.

3.2 Capturing social business process requirements using meta-modeling

In this subsection, we develop the necessary meta-models for capturing SBP requirements from the organizational, technological, and management perspectives.

3.2.1 Requirements from the organizational perspective

SBP requirements from the organizational perspective allow BP engineers to establish the necessary procedures for an efficient use of Web 2.0 technologies and applications within the enterprise. A close examination of the enterprise architecture results into identifying the relevant organizational units that must be adapted to socialization requirements. In our work, we rely on the organizational perspective of the ArchiMate language [21] to identify socialization requirements from an organizational perspective. These re-

quirements must specify when, where, and who may perform social tasks and which data (business objects) can be communicated over Web 2.0 [4]. Our approach allows to capture these requirements thanks to the meta-model of Figure 1 that includes the following meta-classes:

- **Business actor:** An organizational entity (person or a group of persons) that has a status within the enterprise. Adopting an SBP requires the identification of the business actors that will be engaged in the execution of social tasks.
- **Business role:** A specific behaviour of a business actor participating in a particular context. SBP adoption may require the definition of new business roles that will perform social tasks (e.g., social marketing chief).
- **Duties:** A right or obligation associated with a business role and it is required to perform social tasks in an SBP. These duties must be identified to guarantee the ability of a business role to perform a social tasks (e.g., ability to access a Web 2.0 application).
- **Skills:** The necessary competencies that a business role should have to perform social tasks within an SBP. These skills must be identified to assess the readiness of enterprise' employees' socialization (e.g., capability to perform a social task).
- **Business Object:** A relevant unit of information that can be communicated over Web 2.0. These business objects must be identified to prevent violation of enterprise' policies (e.g., security policies).
- **Social application service:** a functionality provided by a Web 2.0 application (e.g., create account, publish, comment, etc).

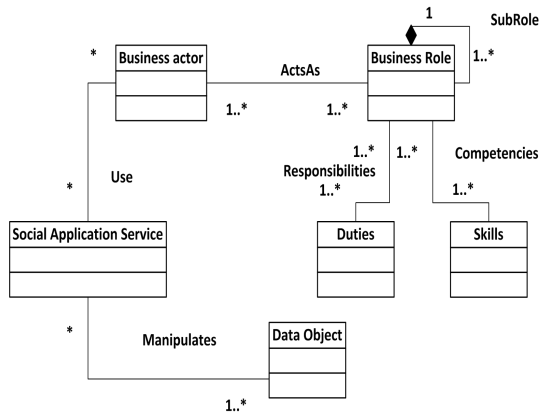


Figure 1: Requirements meta-model from an organizational perspective.

3.2.2 Requirements from a technological perspective

The ArchiMate language describes the hardware and software components used within an enterprise [21]. Based on this description, we focus on the necessary IT components that are required to perform an SBP. From a technological perspective, SBP requirements state the necessary hardware

and software components that guarantee a successful execution of the new social interactions and help enterprises select the most appropriate Web 2.0 technologies/applications according to their missions and goals [4]. It is important to identify the necessary social application services (e.g., profiling, messaging, and voting) which must be used to execute the BP and the infrastructure upon which these services will run [32].

To help BP engineers list the technological requirements, we propose the meta-model depicted in Figure 2 where:

- **Social application service:** A functionality that a Web 2.0 application should provide to execute social tasks. The identification of the social services helps enterprises find the most suitable Web 2.0 application that will be used to perform a SBP.
- **Functional requirement:** Represents the functionalities that the social application service should fulfil; for example, a service must guarantee group video calls. These functionalities are essential for a proper operation of an SBP and thus must be guaranteed by the used service.
- **Non-functional requirement:** Represents the constraints on the functionalities provided by a social application service; for example, a messaging service that is secured during transmission. These non-functional requirements are essential for a proper operation of an SBP and thus must be guaranteed by the used service.
- **Configuration parameter:** Describes how a social application service can be adopted to users' requirements. For example, configure the sharing service to identify who can see and/or share a publication. Configuration parameters may be used to adopt a social application service to SBP requirements.
- **Infrastructure:** Refers to the required hardware and software components for an efficient execution of a social application service. Each infrastructure element (e.g., Internet connection) has a set of characteristics (e.g., bandwidth). The identification of the required infrastructure can help BP engineers assess the infrastructure readiness for socialization.

3.2.3 Requirements from a management perspective

To provide guidelines for assessing the effectiveness of SBPs, we handle socialization requirements from a management perspective. Indeed, it is important to equip enterprises with mechanisms for measuring their return on investment and achievement degree of socialization goals (i.e., why the enterprise embraces Web 2.0) [32]. The proposed approach for capturing requirements from a management perspective considers the defined socialization goals. For each socialization goal, BP engineers should specify the set of KPIs used to measure its achievement degree [14, 25]. Figure 3 describes the management requirements meta-model:

- **Goal:** Final, observable, and measurable result that should be achieved within a certain deadline. In our case, we are interested to socialization goals. These goals are important for the evaluation of socialization benefits.

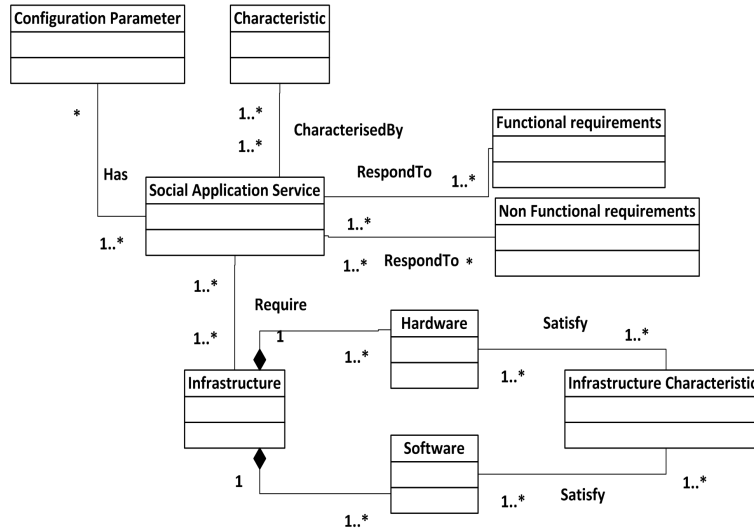


Figure 2: Requirements meta-model from a technological perspective.

- **Key performance indicator:** Key concept to measure the achievement of an enterprise goals. These KPIs are closely linked to socialization goals and should be carefully defined. Thanks to these KPIs the value of return on investment can be estimated.
- **Current Value:** The current value of a KPI that measures the progress towards the satisfaction of a socialization goal.
- **Threshold Value:** The minimum acceptable value that a KPI may have. If the value of a KPI is less than this threshold, the corresponding goal is not satisfied.
- **Target Value:** The value that the KPI should have to assume that the corresponding goal is reached. If it is reached by a KPI, the corresponding goal will be satisfied.
- **KPI monitor:** describes how the different KPIs are monitored.

4. SOCIAL BUSINESS PROCESS

The BP meta-model, proposed by Curtis et al. in [10], encompasses the core BP components which are categorized into four aspects: functional, organizational, behavioral, and informational. The functional aspect focuses on the tasks in a BP. A task is either atomic or composite. The organizational aspect describes the units that participate in the BP execution. The behavioral aspect represents the flows and control nodes in the BP. Finally, the informational aspect describes the entities that a BP produce or manipulate. These entities could be event, information resource, and tangible resource. These four perspectives are commonly modeled using the OMG standard BPMN [24].

In the literature, many studies discuss the benefits of Web 2.0 technologies and applications for businesses [12, 31]. These include communication, collaboration, sharing, and collecting knowledge. To achieve different goals, many

enterprises adopt Web 2.0 applications. For instance, Salesforce.com incorporates some social interactions within some BPs, especially to maintain its employees connected [2].

In a previous work, we define an SBP as “a BP that uses any Web 2.0 technology or application to achieve the enterprise’s business goals and to foster communication, collaboration and exchange of knowledge either among internal or with external actors in everyday work” [36]. In this section, we propose a meta-model based definition for SBP (SBP2M).

Accounting the proposed meta-model for capture SBP requirements, we suggest SBP2M, which is obtained by enriching the existing BP meta-model [10] with the necessary concepts to support the social dimension of an SBP. These new concepts are identified thanks to the proposed meta-models (the grey colored concepts in Figure 4):

- The functional aspect is enriched by social task that is either atomic or composite. A social task is executed by a Web 2.0 application (e.g., posting advertisements on Facebook).
- The organizational aspect is enriched by social actor, which describes those who will perform social tasks. A social actor may be either human (internal or external to the enterprise) or Web 2.0 application.
- The informational aspect is enriched by social concepts like Web 2.0, social event, and social data object. A social task may produce zero or many social events. In addition, a social task manipulates or produces social data object (e.g., advertisement data) that could be made available over Web 2.0.
- The behavioral aspect is enriched by a social decision that is made over Web 2.0 applications and is represented inside the SBP model as a control node. The behavioral aspect is also enriched by a social flow; it describes the interactions that take place between actors over Web 2.0 applications. For example, a social flow can be used to model the link between posting

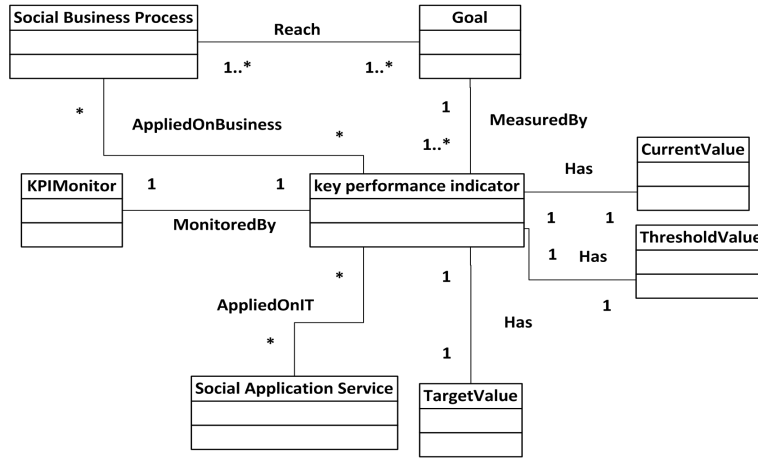


Figure 3: Requirements meta-model from a management perspective.

task and the corresponding commenting and collecting feedbacks tasks.

Thanks to SBP2M, the different business entities that socialization could affect can be explicitly described (e.g., task, event, and information resource). The next section focuses on modeling the socialization requirements, which considers the social dimension of a BP.

5. SOCIAL BUSINESS PROCESS REQUIREMENTS MODELING

BPMN is the OMG standard for BP modelling. Its concepts are organized into four categories: objects flow (e.g., activities, events, and gateways), connecting objects (e.g., sequence flow, message flow, and association), swimlanes (e.g., pools and lanes), and artefacts (e.g., data object, group, and annotation). The OMG maintains BPMN that has recently been adopted as an ISO standard. Indeed, ISO/IEC 19510:2013 provides a notation that is easy to understand and use by users like business analysts [20]. In addition, BPMN syntax is extensible as new markers can be added to BPMN concepts. In this paper, we propose a specific notation to model requirements linked to the SBP model. This notation supports modelling requirements captured through the meta-models depicted in Figure 1, Figure 2, and Figure 3. This extension is defined according to the BPMN extensibility mechanism, which is based on UML class diagram that represents the meta-model of the new extensions. BPMN extension consists essentially of four different elements: Extension, ExtensionDefinitions, Extension-AttributesDefinition, and ExtensionAttributesValue [24].

- ExtensionDefinition is independent of BPMN elements and defines and groups the extension attributes.
- ExtensionAttributeDefinition defines the set of new attributes that can be attached to BPMN element.
- ExtensionAttributeValue represents actual extension attribute value.
- Extension links an ExtensionDefinition and its attributes to a specific BPMN model definition.

Figure 5.a illustrates the socialization requirements extension meta-model class diagram according to BPMN 2.0 extensibility mechanism [24]. SocializationRequirements class defines the socialization requirements extension, which refers to an ExtensionDefinition class. SocializationRequirements class groups the attribute Definition classes of: organizational, technological, and management classes using composite relationship as illustrated in Figure 5.b. Each attribute has a set of attribute values like KPI, level, and Web2Application (Figure 5.b). The proposed extension allows to enrich the SBP model with the set of requirements that should be satisfied before the adoption of SBP.

In terms of modeling, we adopt the “separation of concerns” principle where business and socialization requirements are modeled independently. To do so, the organizational, technological, and management requirements are modeled using “decorated annotations” within the SBP model (Table 1). The new annotations can be related to any element within the SBP model like task, control node, or data object.

To prove the feasibility of the proposed extension, we developed a graphical editor, which extends the open source editor BPMN 2.0 modeler. Each new notation is created as a customization of the standard BPMN 2.0 annotation. The attributes of each new concept are accessible, like standard attributes, via the property sheet once selected.

Table 1: Concrete syntax of the proposed extension.

Socialization requirement	Notation
Organizational requirement	
Technological requirement	
Management requirement	

6. VALIDATION

In this section, we evaluate the proposed meta-models using BWW ontology and we introduce an example to illustrate the use of our BPMN extension.

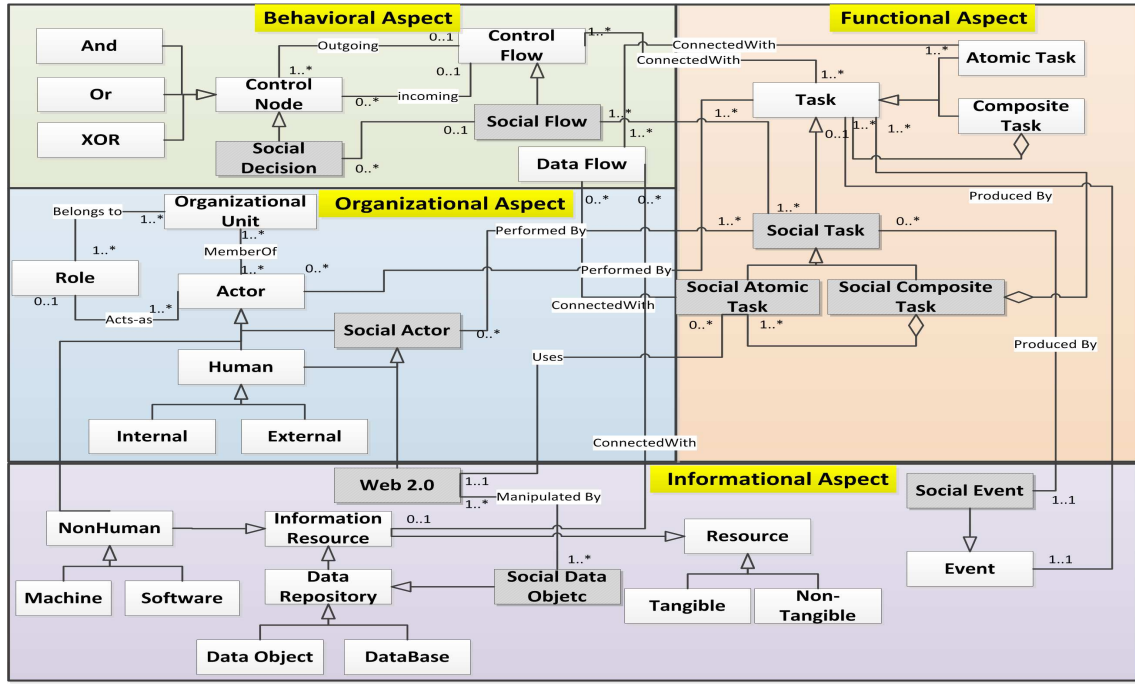


Figure 4: The proposed Social Business Process Meta-Model (SBP2M).

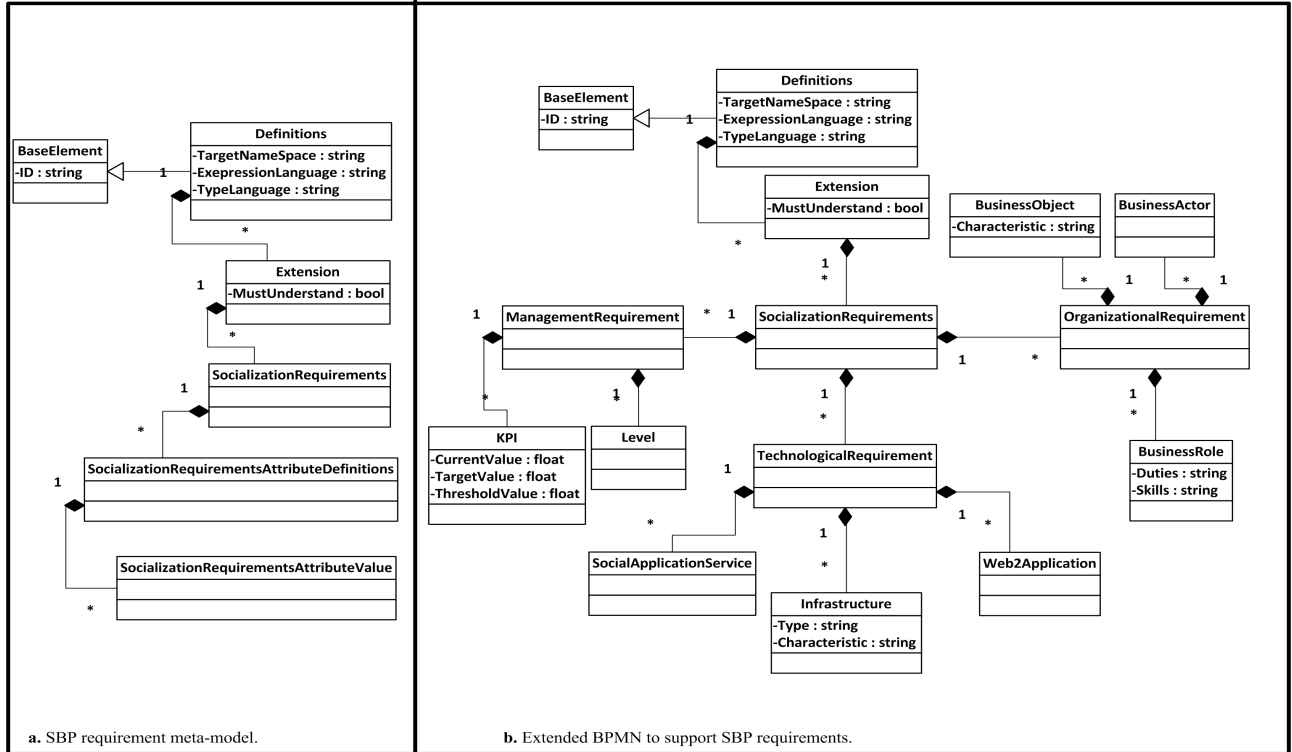


Figure 5: BPMN4SR meta-model: BPMN2.0 extensions with socialization requirements.

6.1 Ontology-based analysis of the proposed meta-models

In this sub-section, we evaluate the suitability of the proposed meta-models to cover SBP requirements. The suit-

ability of the meta-models can be estimated by assessing their completeness and clarity. Completeness measures the ability of the meta-models to cover all the social aspects of an SBP while clarity measures how easy to learn and re-

member the concepts of a modeling languages [18]. In order to estimate the completeness and clarity of the proposed meta-models, we used an ontology based analysis. Indeed, ontology-based analysis is widely reported in the literature as a means to evaluate modeling languages [29, 28, 16]. It is based on mapping the concepts of an ontology onto those of the modeling language meta-models and vice versa. One of the most used one for ontology-based analysis is Bunge-Wand-Weber (BWW) [35]. Different studies have shown the benefits of using BWW to evaluate the completeness and clarity of modeling languages [13, 17]. This ontology is used as a benchmark to evaluate several modeling languages like BPMN [28], EPC [16], and Petri Nets [29]. The method proposed by Wand and Weber consists of two mappings: representation and interpretation [35]. The former maps each construct within the BWW model onto a corresponding construct within the meta-model. As a result, we can assess the degree of ontological completeness and degree of redundancy. The latter maps each construct of the meta-model onto corresponding construct in the BWW model. This mapping determines the potential degree of construct overload and construct excess.

The results of these mappings are used to calculate the values of the four ontological dependencies [35]:

- **Completeness:** measures whether a grammar is able to represent the same information about the real world that can be represented in the BWW ontology. A grammar is complete when each construct from the ontology can be mapped at least on one construct of the grammar.
- **Overload:** measures whether each grammatical construct can be mapped on exactly one ontological constructs. Overload occur if at least one of the grammatical construct can be mapped on more than one ontological construct.
- **Redundancy:** measures whether each ontological construct is represented by exactly one grammatical constructs. Redundancy occur if at least one of the ontological construct can be mapped on more than one grammatical construct.
- **Excess:** measures whether each grammatical construct can be mapped on at least one ontological constructs. Excess occur if at least one of the grammatical construct cannot be mapped on an ontological construct.

The ontological dependencies are a set of ratios calculated based on the mappings results. Table 2 summarizes the results of the ontological analysis to prove the completeness and clarity of SBP2M and the requirements meta-model, which is obtained by combining the organizational, technological, and management meta-models.

From the results in Table 2, we note that the construct deficit in SBP2M is 32.14% which means that its completeness is 67.86%. This value is considered satisfactory in comparison to the scores of other languages [13, 17]. The requirement meta-model also has a good completeness value of 64.29%. For ontological clarity, the value of the construct overload is of 7.69% and excess is of 0% for the SBP2M and redundancy is of 22.30%. Similarly, the requirements meta-model whose construct overload is 3.18% and excess is

6.45% for the SBP2M and the redundancy is 14.56%. Overall, these results argues that SBP2M as well as the proposed meta-model for capturing requirements have an acceptable ontological expressiveness.

6.2 Example

To illustrate the use of our graphical editor and the benefits of the proposed extension, we consider an organization attempting to socialize a course planning BP. In this process, the audience is a critical factor, as it should be taken into account at the different stages of the course preparation. First of all, the company must maximize the learners number to secure better profit. Consequently, an advertisement phase is required at the beginning of the project. Secondly, it is important to customize the learning process according to the audience in terms of goals, content, structure, etc. Therefore, the company must know the audience's learning skills, their jobs and capabilities to access online courses, etc. Finally, the company experts must focus on the course content, which plays a critical role in how the audience learns the material. During the development of the course, the experts must collaboratively focus on how the course should be organized.

However, without using Web 2.0 applications, this process may become difficult due to challenges such as advertisement costs, difficulty to collect learning requirements, etc. In contrast, the use of Web 2.0 applications may simplify this process in different manners. For example, the advertisement phase may be freely accomplished using a social network. In fact, social media are considered as a suitable tool for advertisement. Then, the company can collect online learners' feedback using any Web 2.0 application like Facebook. Finally, the set of available Web 2.0 collaborative-writing tools may play an import role in order to simplify the course development task.

This example shows how a BP can be improved using Web 2.0 in terms of advertisement, online feedback collection and collaborative execution of some tasks. However, the successful socialization of such a process should starts by capturing requirements. First of all, the organizational requirements allows to identify the different actors implicated in the execution of this BP either internal (e.g., actors that will perform advertisement activities and experts) or external (e.g., learners). Once identified, it is important to assess their skills and duties to guarantee their ability to perform social task, especially when we deal with external actors (i.e., learners). Indeed, the socialization of such a BP cannot succeed if the audience is not able to perform social tasks. Then, the technological requirements must be examined to identify the most suitable Web 2.0 application to be used by the SBP. This application can be determined through the identification of the required social application services (e.g., a sharing service for advertisement purposes and a commenting or messaging service to collect learners feedbacks, etc.) and the examination of the available infrastructure. Last but not least, to provide a means for the continuous evaluation of the SBP, it is important to deal with management requirements. To this end, the organization must starts by the identification of socialization goals, which include, in the current example, reduce advertisement costs, improve advertisement and collect learner's feedbacks. For each goal, the organization should specify a set of KPIs allowing to assess the degree of its achievement. In this ex-

Table 2: Results of the ontological evaluation.

	Ontological Completeness	Ontological Clarity		
	Deficit	Redundancy	Overload	Excess
SBP meta-model	32.14 %	22.30 %	7.69 %	0 %
Requirements meta-model	35.71 %	14.56 %	3.18 %	6.45 %

ample, these KPIs include advertisement costs, number of new learners, etc. for each KPIs experts specify how these KPIs can be monitored. The representation of these requirements within the BP model facilitate the experts mission. In fact, with the use of our graphical editor, the model will be clear, as socialization requirements will be modelled as decorated annotation within the SBP model. During the implementation, such an annotated model can help developers to generate policies from these requirements. Figure 6 shows the process of planning an online course annotated with requirements.

7. CONCLUSION

The migration towards SBP is expected to bring benefits to enterprises. However, the lack of guidelines and support for the adoption of SBP could constrain their rolling [4]. Indeed, the adoption of SBP instead of traditional BPs can cause changes in everyday-work (e.g., enable access to Web 2.0 resources and control the use of Web 2.0.). In addition, without a clear and consistent definition of what an SBP is, it is difficult to put such processes in place.

In this paper, we defined a meta-modeling approach for capturing socialization requirements from organizational, technological, and management perspectives. The organizational requirements identify the necessary procedures that need to be put in place so that efficient use of the future Web 2.0 technologies and applications is ensured. The technological requirements identify the future Web 2.0 technologies and applications that will help the enterprise sustain its growth and fulfill its mission. Last but not least, the management requirements identify the relevant metrics (or key performance indicators) that permit to evaluate the efficient use of Web 2.0 applications. We also a meta-model based definition for SBP by enriched the different aspects of the BP meta-model described in [10] with the necessary concepts that capture the social dimension of a SBP.

In term of modeling, we have extended BPMN 2.0 to support the modeling of socialization requirements. Indeed, the captured requirements are introduced within the future SBP as annotations. Finally, the proposed SBP and socialization requirements meta-models are evaluated in terms of completeness and clarity using the BWW ontology.

We are currently working on the transformation of the SBP requirements on a set of policies that help enterprises to adopt an SBP and to guarantee its proper functioning.

8. REFERENCES

- [1] E. P. Baumer, V. Khovanskaya, P. Adams, J. P. Pollak, S. Volda, and G. Gay. Designing for engaging experiences in mobile social-health support systems. *IEEE Pervasive Computing*, (3):32–39, 2013.
- [2] S. Bennett. The benefits of communicating and collaborating in a real-time enterprise social network. In *Professional Communication Conference (IPCC), 2012 IEEE International*, pages 1–4. IEEE, 2012.
- [3] M. Brambilla, P. Fraternali, and C. Vaca. A notation for supporting social business process modeling. In *Business Process Model and Notation*, pages 88–102. Springer, 2011.
- [4] V. Buregio, Z. Maamar, and S. Meira. An architecture and guiding framework for the social enterprise. *Internet Computing, IEEE*, 19(1):64–68, 2015.
- [5] R. Cerenkovs and M. Kirikova. Supporting introduction of social interaction in business processes. In *Perspectives in Business Informatics Research*, pages 187–201. Springer, 2014.
- [6] M. A. Chaâbane, E. Andonoff, R. Bouaziz, and L. Bouzguenda. Versions to address business process flexibility issue. In *Advances in Databases and Information Systems*, pages 2–14. Springer, 2009.
- [7] S. Chandler. Social bpm: Gateway to enhanced process efficiency - virtusa official blog. <http://www.virtusa.com/blog/2011/11/social-bpm-gateway-to-enhanced-process-efficiency/>, November 2011. (Accessed on 03/09/2015).
- [8] A. Cockburn. Writing effective use cases. *preparation for Addison-Wesley Longman*. www.infor.uva.es/~mlaguna/is2/materiales/BookDraft1.pdf, 1999.
- [9] R. Cross, P. Gray, S. Cunningham, M. Showers, and R. J. Thomas. The collaborative organization: how to make employee networks really work. *Image*, 2013.
- [10] B. Curtis, M. I. Kellner, and J. Over. Process modeling. *Communications of the ACM*, 35(9):75–90, 1992.
- [11] F. Ellouze, M. Chaabane, R. Bouaziz, and E. Andonoff. Addressing inter-organisational process flexibility using versions: The vp2m approach. In *Research Challenges in Information Science (RCIS), 2016 IEEE Tenth International Conference on*, pages 1–12. IEEE, 2016.
- [12] S. Erol, M. Granitzer, S. Happ, S. Jantunen, B. Jennings, P. Johannesson, A. Koschmider, S. Nurcan, D. Rossi, and R. Schmidt. Combining bpm and social software: contradiction or chance? *Journal of software maintenance and evolution: research and practice*, 22(6-7):449–476, 2010.
- [13] J. Evermann and Y. Wand. Towards ontologically based semantics for uml constructs. In *Conceptual Modeling—ÅTTER 2001*, pages 354–367. Springer, 2001.
- [14] E. Ganesan and R. Paturi. Key performance indicators framework-a method to track business objectives, link business strategy to processes and detail importance of key performance indicators in enterprise business architecture. *AMCIS 2009 Proceedings*, page 736, 2009.

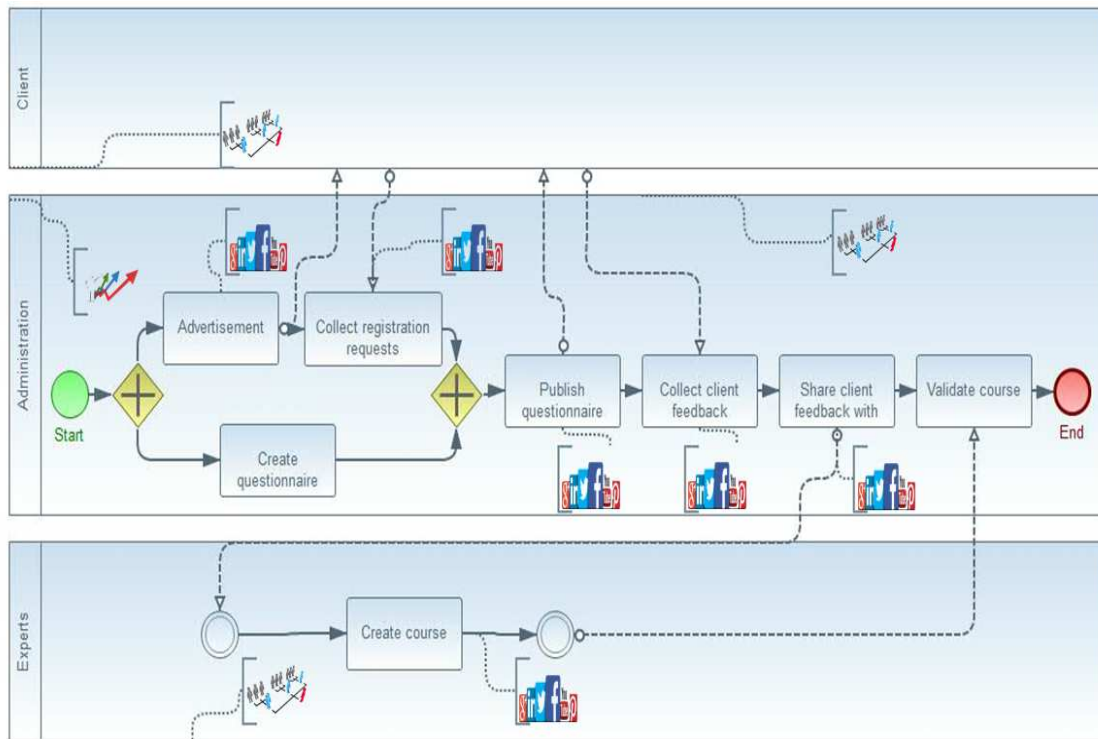


Figure 6: Online course planning process modelled annotated with socialization requirements.

- [15] A. Goknil, I. Kurtev, and K. van den Berg. A metamodeling approach for reasoning about requirements. In *Model Driven Architecture–Foundations and Applications*, pages 310–325. Springer, 2008.
- [16] P. Green and M. Rosemann. Integrated process modeling: an ontological evaluation. *Information systems*, 25(2):73–87, 2000.
- [17] F. Heidari, P. Loucopoulos, F. Brazier, and J. Barjis. A meta-meta-model for seven business process modeling languages. In *Business Informatics (CBI), 2013 IEEE 15th Conference on*, pages 216–221. IEEE, 2013.
- [18] P. v. B. S. Hoppenbrouwers, S. O. E. Proper, and J. Barjis. The practice of enterprise modeling. 2010.
- [19] IBM. Applying social business: The repeatable patterns that improve business processes and provide return. <http://www-01.ibm.com/common/ssi/cgi-bin/ssialiasinfotype=SAsubtype=WHhtmlfid=EPW14036USEN>, January 2014. (Accessed on 01/02/2016).
- [20] ISO. Iso/iec 19510:2013 information technology object management group business process model and notation. http://www.iso.org/iso/catalogue_detail.htm?csnumber=62652, July 2013. (Accessed on 02/01/2016).
- [21] M. Lankhorst. *Enterprise architecture at work : modelling, communication and analysis*. The enterprise engineering series. Springer, Heidelberg, New York, 2013. lccopycat.
- [22] B. List and B. Korherr. An evaluation of conceptual business process modelling languages. In *Proceedings of the 2006 ACM symposium on Applied computing*, pages 1532–1539. ACM, 2006.
- [23] A. P. McAfee. Enterprise 2.0: The dawn of emergent collaboration (2006), 2008.
- [24] OMG. Bpmn 2.0. <http://www.omg.org/spec/BPMN/2.0/>, January 2011. (Accessed on 02/07/2015).
- [25] D. Parmenter. *Key performance indicators: developing, implementing, and using winning KPIs*. John Wiley & Sons, 2015.
- [26] N. Ramadan and N. Al-Qirim. New hybrid web 2.0 adoption governance framework for public sector. *International Journal of e-Collaboration (IJeC)*, 11(1):8–21, 2015.
- [27] A. Rashid, A. Moreira, and J. Araújo. Modularisation and composition of aspectual requirements. In *Proceedings of the 2nd international conference on Aspect-oriented software development*, pages 11–20. ACM, 2003.
- [28] J. Recker, M. Rosemann, M. Indulska, and P. Green. Business process modeling—a comparative analysis. *Journal of the Association for Information Systems*, 10(4):1, 2009.
- [29] J. C. Recker and M. Indulska. An ontology-based evaluation of process modeling with petri nets. *IBIS-International Journal of Interoperability in Business Information Systems*, 2(1):45–64, 2007.
- [30] I. Roussaki, N. Kalatzis, N. Liampotis, P. Kosmides, M. Anagnostou, K. Doolin, E. Jennings, Y. Bouloudis, and S. Xynogalas. Context-awareness in wireless and

mobile computing revisited to embrace social networking. *Communications Magazine, IEEE*, 50(6):74–81, 2012.

- [31] R. Schmidt and S. Nurcan. Bpm and social software. In *Business Process Management Workshops*, pages 649–658. Springer, 2009.
- [32] U. Sivarajah, Z. Irani, and V. Weerakkody. Evaluating the use and impact of web 2.0 technologies in local government. *Government Information Quarterly*, 32(4):473–487, 2015.
- [33] R. Soley et al. Model driven architecture. *OMG white paper*, 308(308):5, 2000.
- [34] A. Van Lamsweerde. Goal-oriented requirements engineering: a roundtrip from research to practice [engineering read engineering]. In *Requirements Engineering Conference, 2004. Proceedings. 12th IEEE International*, pages 4–7. IEEE, 2004.
- [35] Y. Wand and R. Weber. On the ontological expressiveness of information systems analysis and design grammars. *Information Systems Journal*, 3(4):217–237, 1993.
- [36] F. Yahya, K. Boukadi, Z. Maamar, and H. Ben-Abdallah. Enhancing business processes with web 2.0 features. In *ICE-B 2015 - Proceedings of the 12th International Conference on e-Business, Colmar, Alsace, France, 20-22 July, 2015.*, pages 183–190, 2015.